

Quantum Computing — Fundamental Analysis

75+ companies and entities ranked by irreplaceability in the quantum value chain. From qubit fabrication to quantum-safe cryptography — who are the toll booths of the quantum era?

Analysis date: August 6, 2026 · Investment horizon: 40 years · Framework: §64–§67

COMPANIES ANALYZED	TIER 1 — IRREPLACEABLE	TIER 2 — CRITICAL	TIER 3 — IMPORTANT	TIER 4 — RELEVANT	INDUSTRY STAGE
78	8	16	24	30	Pre-Revenue
Listed + key private	Quantum halts without them	Very hard to substitute	Strong but substitutable	Positioned for quantum era	Most pure-plays unprofitable

Important Context: Quantum Computing Is Pre-Paradigmatic

Unlike semiconductors (a \$600B mature industry), quantum computing is **pre-revenue for most pure-plays**. The industry is where semiconductors were in the 1960s — foundational physics is still being debated (superconducting vs trapped ion vs photonic vs topological). This changes the irreplaceability analysis fundamentally:

- **Hardware approach is NOT settled** — a company irreplaceable today may be irrelevant if its qubit modality loses. Diversification across modalities is critical.
- **The real toll booths are enabling infrastructure** — dilution refrigerators, control electronics, EDA for quantum, error correction IP — these win regardless of which qubit type prevails.
- **Classical companies with quantum divisions** (IBM, Google, Microsoft) are the safest exposure — quantum optionality on top of profitable core businesses.
- **Financial health matters more here** — pure-play quantum companies burn cash. Many won't survive to commercialization. Balance sheet = survival probability.

Irreplaceability Score weights: Technology uniqueness (30%) · Modality independence (20%) · IP/patent depth (20%) · Supply chain criticality (15%) · Ecosystem lock-in (15%)

TIER 1 Irreplaceable — Quantum Cannot Advance Without Them

These 8 companies/divisions provide capabilities that no other entity on Earth can replicate within 5 years. Most are enabling infrastructure — they win regardless of which qubit modality prevails. The true toll booths of quantum.

#	COMPANY	SEGMENT	MKT CAP	REV (TTM)	GROSS %	OP %	P/E	D/E	CASH	BURN RATE	HEALTH	IRREPL.	MOATS
1	Bluefors Private (Finland)	Dilution Refrigerators	Private	~\$200M	~55%	~25%	—	Low	—	Profitable	STRONG	98	Near-Monopoly IP
~70% of all dilution refrigerators for quantum computing. Every superconducting quantum computer (IBM, Google, Amazon) requires cooling to 15 millikelvin. Without Bluefors, no superconducting qubits. Not directly investable — watch for IPO. The most irreplaceable company in quantum.													
2	IBM IBM	Full-Stack Quantum	\$220B	\$62B	56%	17%	22x	2.30	\$13B	Profitable	MODERATE	95	Ecosystem IP Scale
Most advanced quantum program: 1,000+ qubit processors (Condor/Heron), Qiskit is the de facto standard SDK (~500K users), quantum-centric supercomputing roadmap through 2033. 2,000+ quantum patents. D/E high but quantum is optionality on top of \$62B hybrid cloud business. The "IBM of quantum" — literally.													
3	Alphabet (Google) GOOGL	Quantum AI Lab	\$2,200B	\$340B	57%	30%	22x	0.05	\$100B+	Profitable	STRONG	94	IP Talent Scale
Willow chip achieved "below-threshold" quantum error correction (2024) — the biggest quantum milestone since Shor's algorithm. Quantum supremacy (2019). Owns Cirq SDK. Unlimited R&D budget. Google's quantum team (Hartmut Neven, John Martinis alumni) is the world's deepest talent pool. Pure quantum optionality on the world's best balance sheet.													

#	Company	Segment	Mkt Cap	Rev (TTM)	Gross %	Op %	P/E	D/E	Cash	Burn Rate	Health	Irrepl.	Moats
4	Microsoft MSFT	Topological/Azure Quantum	\$3,200B	\$250B	69%	44%	34x	0.25	\$75B	Profitable	STRONG	92	PlatformIPEcosystem
Azure Quantum is the leading quantum cloud platform (hosts IonQ, Quantinuum, Rigetti). Pursuing topological qubits (Majorana) — if successful, would leapfrog all competitors in error rates. Q# language. Acquired quantum IP from Station Q (UCSB). The quantum cloud infrastructure layer is MSFT's toll booth.													
5	Quantinuum Private (Honeywell spin)	Trapped-Ion Full-Stack	~\$5B (last round)	~\$100M	~40%	Negative	—	Low	~\$500M	~\$150M/yr	GOOD	93	IPFidelityVertical
Highest-fidelity quantum computer in the world (H2 processor, 99.9%+ two-qubit gates). Honeywell backing provides manufacturing discipline. TKET compiler widely used. Trapped-ion approach may prove superior for error correction. IPO expected 2026–2027. Invest via Honeywell (HON) for now — 54% ownership.													
6	Honeywell HON	Quantinuum Parent + Quantum Infra	\$145B	\$37B	37%	22%	22x	0.85	\$10B	Profitable	GOOD	88	OwnershipManufacturing
54% owner of Quantinuum. Also provides precision manufacturing (vacuum chambers, ion traps) that is directly transferable to quantum hardware. Best listed proxy for pure-play quantum. Industrial conglomerate with quantum optionality — passes the 5-year test (\$63).													
7	Zurich Instruments Private (acquired by Rohde & Schwarz)	Quantum Control Electronics	Private	~\$80M	~60%	~20%	—	—	—	Profitable	GOOD	92	Near-MonopolyIP
Dominant in quantum control instrumentation (HDAWG, SHFQC). Every qubit needs precise microwave pulses — Zurich Instruments provides the "nervous system." Used by Google, IBM, ETH, MIT. Acquired by Rohde & Schwarz (private). Invest via Keysight (KEYS) as closest listed competitor.													
8	Keysight Technologies KEYS	Test & Measurement / Quantum Control	\$32B	\$5.0B	64%	22%	26x	0.35	\$2B	Profitable	STRONG	86	InstrumentsIPEcosystem
World leader in electronic test & measurement. Quantum control instruments, arbitrary waveform generators, signal analyzers critical for qubit characterization. Every quantum lab uses Keysight. Profitable diversified business with quantum tailwind. Best listed "picks and shovels" play.													

TIER 2

Critical — Leading Platforms & Essential Enablers

The frontrunners in quantum hardware, key enabling technologies, and companies providing infrastructure that all quantum approaches need. Substitution possible but would set the industry back 2–5 years.

#	Company	Segment	Mkt Cap	Rev	P/E	D/E	Cash	Burn	Health	Irrepl.
9	IonQ IONQ	Trapped-Ion Quantum	\$8B	\$45M	N/A	0.10	\$450M	~\$120M/yr	MODERATE	82
Leading listed pure-play quantum computer company. Trapped-ion approach with #AQ (algorithmic qubits) metric. US government contracts (DoD, DOE). Cloud access via AWS/Azure/GCP. ~3.5 years of cash runway. Highest revenue among pure-plays. Commercial quantum networking R&D.										
10	PsiQuantum PSQT (IPO 2025)	Photonic Quantum	\$4B	~\$10M	N/A	0.00	~\$600M	~\$100M/yr	MODERATE	83
Photonic approach using silicon photonics — can leverage existing semiconductor fabs (GlobalFoundries partnership). Room-temperature operation (no dilution fridge). If photonic qubits work at scale, PsiQuantum bypasses the cooling bottleneck entirely. \$750M government backing from Australia and US. High risk but unique manufacturing thesis.										
11	Rigetti Computing RGTI	Superconducting Quantum	\$3B	\$15M	N/A	0.15	\$200M	~\$80M/yr	MODERATE	76
Only pure-play with its own quantum chip fab (Fab-1 in Fremont). Vertically integrated superconducting qubit design + manufacturing. Ankaa-3 processor. Hybrid classical-quantum cloud. UK government partnership. ~2.5 years runway — needs more capital or revenue.										
12	D-Wave Quantum QBTS	Quantum Annealing + Gate	\$2.5B	\$12M	N/A	0.30	\$100M	~\$80M/yr	WEAK	74

#	COMPANY	SEGMENT	MKT CAP	REV	P/E	D/E	CASH	BURN	HEALTH	IRREPL.
	Pioneer in quantum annealing — the only commercially deployed quantum technology solving real optimization problems. 5,000+ qubit Advantage2 system. Now building gate-based systems too. Cash runway is the major risk (~1.2 years). Technology valuable, survival uncertain.									
13	Amazon (AWS Braket + Center for QC) AMZN	Quantum Cloud / Cat Qubits	\$2,100B	\$620B	35x	0.30	\$80B	Profitable	STRONG	82
	AWS Braket hosts IonQ, Rigetti, QuEra, OQC — the marketplace for quantum access. Also building own quantum hardware (cat qubits at AWS Center for Quantum Computing, Caltech). The cloud distribution layer for quantum = platform toll booth.									
14	Linde plc LIN	Helium-3 / Cryogenics	\$220B	\$33B	30x	0.35	\$5B	Profitable	STRONG	84
	World's largest supplier of helium (including rare He-3 isotope critical for dilution refrigerators). Also supplies ultra-pure gases for qubit fabrication. Helium supply is finite and constrained — geopolitical chokepoint. The "Shin-Etsu of quantum."									
15	FormFactor FORM	Cryogenic Probe/Test	\$3B	\$700M	25x	0.05	\$150M	Profitable	GOOD	80
	Leading cryogenic wafer probing — you must test qubits at millikelvin temperatures before deploying them. FormFactor's cryo probers are used by IBM, Google, Intel for qubit characterization. Also a semiconductor play (Tier 3 in semi analysis). Dual exposure.									
16	Nvidia NVDA	Quantum-Classical Hybrid (cuQuantum)	\$3,200B	\$130B	55x	0.40	\$30B	Profitable	STRONG	81
	cuQuantum SDK for GPU-accelerated quantum circuit simulation. DGX Quantum platform (with Quantum Machines). Every quantum researcher uses GPUs for simulation. When quantum goes hybrid, NVIDIA GPUs are the classical half. The bridge between classical and quantum worlds.									
17	Quantum Machines Private (Israel)	Quantum Control (OPX+)	~\$500M	~\$30M	N/A	—	~\$100M	~\$25M/yr	GOOD	85
	OPX+ quantum controller is becoming the industry standard for qubit control. Partnership with NVIDIA (DGX Quantum). Real-time, low-latency feedback loops for error correction. The "Zurich Instruments competitor" that's gaining share rapidly. Not listed.									
18	QuEra Computing Private	Neutral-Atom Quantum	~\$1B	Minimal	N/A	—	~\$200M	~\$40M/yr	GOOD	82
	Harvard/MIT spinout pioneering neutral-atom approach — potentially the most scalable qubit technology (thousands of qubits arranged by laser tweezers). DARPA contracts. AWS Braket access. The dark horse modality that could win the race. Not listed — IPO potential.									
19	Thales HO.PA	Quantum-Safe Crypto / QKD	\$45B	\$22B	20x	0.45	\$5B	Profitable	STRONG	78
	World leader in HSMs (hardware security modules) and encryption infrastructure. When quantum breaks RSA, every government and bank needs quantum-safe encryption. Thales's existing HSM install base is the upgrade path. Also developing QKD (quantum key distribution). Defense/aerospace diversification.									
20	Palo Alto Networks PANW	Post-Quantum Security	\$130B	\$8B	50x	0.60	\$3B	Profitable	GOOD	72
	Leading cybersecurity platform positioning for post-quantum cryptography (PQC). Every firewall, VPN, and endpoint needs PQC migration before "Q-Day." Massive installed base that will need quantum-safe upgrades. Not pure quantum but the cybersecurity toll booth for the quantum era.									
21	Coherent Corp COHR	Photonics / Lasers	\$15B	\$5B	35x	0.80	\$1B	Profitable	MODERATE	77
	Precision lasers critical for trapped-ion (qubit manipulation) and neutral-atom (optical tweezers) quantum computers. Also SiC substrates and optical networking. Multi-angle quantum exposure across different modalities. D/E from II-VI merger is deleveraging.									
22	IPG Photonics IPGP	Fiber Lasers	\$5B	\$1.1B	22x	0.05	\$500M	Profitable	STRONG	73
	World leader in high-power fiber lasers. Critical for neutral-atom quantum (optical trapping). Also photonic quantum computing enabler. Near-zero debt, strong balance sheet. Quantum is small but growing use case on top of industrial laser dominance.									

#	COMPANY	SEGMENT	MKT CAP	REV	P/E	D/E	CASH	BURN	HEALTH	IRREPL.
23	National Instruments (now part of Emerson) EMR	Lab Instrumentation	\$65B	\$17B	25x	0.50	\$2B	Profitable	STRONG	72
NI (acquired by Emerson) provides LabVIEW, PXI platforms, and modular instruments used in every quantum lab worldwide. The "operating system" of quantum research instrumentation. Invest via Emerson (EMR).										
24	Photon Spot Private	Single-Photon Detectors	Private	~\$10M	—	—	—	—	N/A	84
Superconducting nanowire single-photon detectors (SNSPDs) — critical for photonic quantum computing, QKD, and quantum networking. Near-monopoly in highest-performance SNSPDs. Tiny company, huge criticality. Not investable directly.										

TIER 3

Important — Strong Positions, Alternatives Exist

Companies with meaningful quantum programs or enabling technologies where competitors or alternative approaches exist. Many are excellent investments because they offer quantum upside on top of profitable core businesses.

#	COMPANY	SEGMENT	MKT CAP	P/E	D/E	HEALTH	IRREPL.
25	Intel INTC	Silicon Spin Qubits	\$90B	N/A	0.50	WEAK	72
Tunnel Falls chip: silicon spin qubits manufactured on existing CMOS fabs. If silicon qubits work, Intel's fab infrastructure gives it unmatched scalability. Quantum is a lottery ticket on a struggling company — but the manufacturing thesis is compelling.							
26	IQM Quantum Private (Finland)	Superconducting (Europe)	~\$1B	N/A	—	MODERATE	70
Europe's leading superconducting quantum computer maker. VTT Finland partnership. On-premise quantum computers for government/defense. Strategic European sovereign quantum play. Not listed.							
27	Oxford Instruments OXIG.L	Cryogenics / Nanofab Tools	\$2.5B	30x	0.10	STRONG	76
Dilution refrigerators (ProtoStar competitor to Bluefors), nanofabrication tools, and cryogenic systems. #2 in quantum cryogenics. Also makes the etching/deposition tools for qubit fabrication. UK-listed, clean balance sheet. Best listed cryogenics play.							
28	Analog Devices ADI	Precision Analog for Qubit Control	\$110B	32x	0.20	STRONG	70
Precision DACs, ADCs, and mixed-signal ICs for quantum control electronics. The analog-to-digital interface at millikelvin temperatures. Also a Tier 2 semiconductor play — dual exposure. Quantum is a small but growing market for ADI's precision products.							
29	Infleqtion (formerly ColdQuanta) Private	Neutral-Atom / Quantum Sensors	~\$500M	N/A	—	MODERATE	73
Neutral-atom quantum computing + quantum sensors (atomic clocks, inertial navigation). DARPA contracts. Quantum sensing may reach commercialization before quantum computing. Multi-product quantum company. Not listed.							
30	Xanadu Private (Canada)	Photonic Quantum / PennyLane	~\$500M	Minimal	—	MODERATE	72
Photonic quantum computing with squeezed light. PennyLane is the leading open-source quantum ML framework (cross-platform, works with any hardware). PennyLane's ecosystem moat may outlast the hardware bet. Not listed.							
31	Atom Computing Private	Neutral-Atom Quantum	~\$500M	Minimal	—	MODERATE	70
1,200+ qubit neutral-atom system demonstrated. Acquired by Renesas investor network adjacent. Nuclear-spin qubits with long coherence times. Competitive with QuEra in neutral-atom race. Not listed.							
32	Corning GLW	Specialty Glass / Fiber Optic	\$42B	22x	0.55	GOOD	68
Optical fiber for quantum networks, specialty glass for photonic circuits. When quantum networking scales, Corning's fiber infrastructure is the physical layer. Also critical for classical AI data center interconnect. Quantum is long-term optionality.							
33	NVIDIA	See Tier 2 #16 — cuQuantum and DGX Quantum platform. Listed once for primary quantum role.					
34	CrowdStrike CRWD	Post-Quantum Endpoint Security	\$80B	60x	0.20	GOOD	65
Endpoint security leader that will need PQC migration. Every enterprise endpoint is a PQC upgrade opportunity. Not a quantum company per se, but the "harvest now, decrypt later" threat makes PQC a near-term revenue driver.							
35	Entrust (Datacard) Private	PKI / Quantum-Safe Certificates	Private	~\$1B	—	N/A	72

#	COMPANY	SEGMENT	MKT CAP	P/E	D/E	HEALTH	IRREPL.
	Major PKI/certificate authority provider. Every TLS certificate needs PQC migration before Q-Day. Entrust's installed base across governments and banks is the upgrade path. Not listed (owned by Thoma Bravo).						
36	Arqit Quantum ARQQ	Quantum Encryption (Software)	\$150M	N/A	0.05	WEAK	60
	QuantumCloud platform for quantum-safe encryption key agreement. UK government and NATO contracts. Pivoted from satellite QKD to software-based approach. Very small, speculative, but positioned in post-quantum security niche.						
37	Toshiba 6502.T	QKD / Quantum Communication	\$18B	12x	0.60	MODERATE	68
	World leader in quantum key distribution (QKD) — deployed commercially in London, Tokyo. Only company with production QKD systems. Multiplexed QKD over existing fiber. Quantum communication may commercialize before quantum computing.						
38	ID Quantique Private (Switzerland)	QRNG / QKD	Private	~\$30M	—	N/A	74
	Pioneer in quantum random number generators (QRNG) — chips shipping in Samsung phones. Also QKD systems. The only quantum technology already in mass consumer devices. SK Telecom is majority owner. Not directly listed.						
39	Pasqal Private (France)	Neutral-Atom Quantum	~\$500M	Minimal	—	MODERATE	70
	European neutral-atom leader (Palaiseau, France). Founded by Alain Aspect (Nobel Prize 2022 for quantum entanglement). 1,000+ qubit roadmap. DARPA and French military contracts. EU's answer to QuEra. Not listed.						
40	Quantum Computing Inc QUBT	Photonic Optimization	\$800M	N/A	0.20	WEAK	55
	Photonic quantum optimization (Dirac-3 entropy computing). Acquired thin-film lithium niobate (TFLN) manufacturing capability from Turbine. Pivot to photonic chips. Speculative but TFLN technology is valuable regardless of quantum outcome.						
41	SandboxAQ Private (Google spinout)	AI + Quantum Security	~\$5B	~\$100M	—	N/A	73
	Google spinout combining AI and quantum. AQ Decrypt for cryptographic vulnerability discovery. Sells to US government and enterprises. Eric Schmidt-backed. Revenue-generating at scale. IPO potential. The commercial PQC migration leader.						
42	Multiverse Computing Private (Spain/EU)	Quantum Finance Software	~\$100M	~\$5M	—	MODERATE	62
	Spanish quantum software company focused on finance (portfolio optimization, risk). Compresión tensorial approach works on classical hardware today. BBVA, CaixaBank clients. European quantum software champion. Tiny but relevant for Spanish investor interest.						
43	Zapata AI ZPTA (merged)	Quantum-Classical AI Software	\$50M	N/A	High	WEAK	50
	Quantum-inspired algorithms for enterprise AI. Financial stress — merged/restructured. Technology (Orchestra platform) has value but survival uncertain. Speculative micro-cap.						
44	Classiq Technologies Private (Israel)	Quantum EDA / Compiler	~\$300M	~\$10M	—	N/A	74
	High-level quantum circuit synthesis — the "Synopsis of quantum." Instead of hand-coding quantum circuits, Classiq auto-compiles from high-level descriptions. If quantum scales, circuit design tools become essential. The EDA parallel is compelling. Not listed.						
45	Strangeworks Private	Quantum Orchestration Platform	~\$100M	~\$5M	—	MODERATE	60
	Quantum computing orchestration platform — hardware-agnostic access to multiple quantum backends. The "middleware" layer. Partnerships with IBM, AWS, Azure. Small but positioned as infrastructure layer.						
46	QC Ware Private	Quantum Algorithms / Enterprise SW	~\$100M	~\$5M	—	MODERATE	58
	Enterprise quantum software — algorithms for chemistry, optimization, ML. Goldman Sachs, Airbus, BMW clients. Strong algorithm IP. Acquired by Nvidia-backed consortium. Integration with classical AI workflows.						
47	Fujitsu 6702.T	Quantum-Inspired / Superconducting	\$30B	12x	0.25	GOOD	65
	Digital annealer (quantum-inspired) already commercially deployed. Superconducting 64-qubit processor in development with RIKEN. Japan's leading quantum program. Enterprise IT base provides distribution channel.						
48	NEC Corporation 6701.T	Quantum Annealing / Vector Engine	\$18B	14x	0.35	GOOD	62
	Quantum annealing and simulated annealing hardware (SX-Aurora). Strong in quantum-inspired optimization for logistics/supply chain. Japanese enterprise base. Quantum is a small division but strategically positioned.						

Companies with quantum exposure through materials, components, software, or strategic positioning. Many are not "quantum companies" per se but will benefit from quantum commercialization through existing product lines or market positions.

#	COMPANY	SEGMENT	MKT CAP	P/E	HEALTH	IRREPL.
49	Applied Materials AMAT	Qubit Fabrication Equipment	\$145B	22x	STRONG	65
	Deposition and etch tools used to fabricate superconducting qubit circuits (Josephson junctions). Also for silicon spin qubit fabrication. When quantum scales, AMAT's existing tools are the manufacturing base.					
50	Lam Research LRCX	Qubit Fabrication Equipment	\$110B	25x	STRONG	62
	Etch tools for qubit fabrication. Less directly relevant than AMAT for current quantum but critical if quantum chip manufacturing scales to high volumes.					
51	ASML ASML	Lithography (Future Quantum Fabs)	\$340B	32x	STRONG	58
	Current quantum chips don't need EUV, but silicon spin qubits (Intel's approach) would leverage standard CMOS lithography at scale. If Intel's bet works, ASML becomes quantum-relevant. Mostly a semiconductor Tier 1 play.					
52	Air Liquide AI.PA	Helium / Cryogenics	\$100B	28x	STRONG	66
	Duopoly with Linde in industrial gases including helium for quantum cryogenics. Also develops quantum-specific cryogenic solutions. European alternative to Linde for quantum gas supply.					
53	II-VI / Coherent	See Tier 2 #21 — II-VI merged into Coherent. Laser and photonic components for quantum.				
54	Thorlabs Private	Optical Components	Private	—	N/A	68
	The "Amazon of photonics" — optical tables, lenses, fiber optics, mounts used in every quantum lab. Not directly investable. Revenue ~\$1B. Every quantum experiment uses Thorlabs components. The Digi-Key of quantum.					
55	Hamamatsu Photonics 6965.T	Single-Photon Detectors	\$5B	20x	STRONG	70
	Single-photon avalanche diodes (SPADs) and photomultipliers for quantum optics. Japanese precision. Also in semiconductor Tier 3. Dual quantum + semiconductor exposure. Listed, clean balance sheet.					
56	Fortinet FTNT	Post-Quantum Network Security	\$75B	45x	STRONG	60
	Network security with PQC-ready FortiGate firewalls. Every enterprise firewall needs PQC upgrade. Large installed base = recurring upgrade revenue. Not a quantum company but a Q-Day beneficiary.					
57	DigiCert Private	Quantum-Safe Certificates	Private	—	N/A	65
	Largest TLS certificate authority. Every website certificate must migrate to PQC algorithms. DigiCert's PQC-ready certificates are already being tested. Not listed (Thoma Bravo/Clearlake).					
58	Booz Allen Hamilton BAH	Quantum Consulting / Gov Integration	\$20B	25x	GOOD	58
	Leading US government tech consultant with quantum practice. Advises DoD, NSA, DOE on quantum strategy. When agencies adopt quantum, Booz Allen integrates it. The "Accenture of quantum for government."					
59	Northrop Grumman NOC	Quantum Sensing / Defense	\$80B	18x	GOOD	58
	Quantum sensing for navigation (GPS-denied environments), quantum networking for secure defense communications. DARPA quantum contracts. Defense provides the first paying quantum customers.					
60	Lockheed Martin LMT	Quantum Computing Integration	\$130B	18x	GOOD	55
	First D-Wave customer. Quantum computing for complex simulation (aerodynamics, cryptography). Defense budget insulates from commercial cycles. Early adopter positioning for quantum advantage in defense applications.					
61	Raytheon (RTX) RTX	Quantum Sensing / PQC Defense	\$165B	22x	GOOD	55
	BBN Technologies division pioneered quantum networking (first quantum internet demo). Quantum radar, quantum sensing for missiles. Collins Aerospace for quantum-safe avionics communications.					
62	Samsung 005930.KS	Quantum Dot Displays / Memory	\$350B	12x	GOOD	52
	Quantum-dot display technology (not quantum computing). Also investing in quantum computing R&D (silicon spin qubits). ID Quantique QRNG in Samsung Galaxy phones. Broad tech conglomerate with quantum touchpoints.					
63	SK Telecom 017670.KS	Quantum Cryptography / ID Quantique	\$12B	10x	GOOD	62
	Majority owner of ID Quantique (QRNG chips, QKD). Deploying quantum-safe telecom network in Korea. Best listed proxy for ID Quantique's quantum technology. Cheap at 10x P/E.					

#	COMPANY	SEGMENT	MKT CAP	P/E	HEALTH	IRREPL.
64	BT Group BT.A.L	Quantum Network Testbed	\$18B	8x	MODERATE	52
Operating UK's quantum-secured metro network (Toshiba QKD). First telecom to deploy commercial quantum key distribution. When quantum networks scale, telcos are the physical infrastructure. Very cheap at 8x.						
65	Deutsche Telekom DTE.DE	Quantum Network (EuroQCI)	\$160B	14x	GOOD	52
Leading participant in EuroQCI (European quantum communication infrastructure). Open Quantum Networking program. When EU deploys sovereign quantum networks, DT's infrastructure is the backbone.						
66	Telefónica TEF	Quantum Network (Spain/EU)	\$25B	12x	MODERATE	50
Spanish quantum network deployment (MadQCI — Madrid quantum communication infrastructure). Partnership with UPM. EuroQCI participant. Relevant for Spanish investor — local quantum infrastructure play. D/E elevated but manageable.						
67	Palantir PLTR	Quantum-Classical Data Integration	\$150B	100x+	GOOD	50
Data integration platform that will likely be among first to integrate quantum computation for government/defense analytics. Not a quantum company but a distribution channel for quantum results. Premium valuation.						
68–78	Also tracked: Atos/Eviden (EU quantum emulator, myQLM — restructuring), OQC (Oxford Quantum Circuits, UK superconducting — private), Alice & Bob (cat qubits, France — private), Diraq (silicon spin qubits, Australia — private), SEEQC (digital quantum, SFQ control — private), Bleximo (application-specific quantum — private), QunaSys (quantum chemistry software, Japan — private), 1QBit (quantum software, Canada — private), ORCA Computing (photonic, UK — private), Nu Quantum (photonic networking, UK — private), Horizon Quantum Computing (quantum compiler, Singapore — private).					

Key Investment Takeaways

Quantum computing investment requires a fundamentally different approach than semiconductors. The industry is pre-revenue, the winning technology is unknown, and most pure-plays will not survive. The 40-year framework (\$64–\$67) actually favors quantum — but only with the right vehicle selection.

The Safest Quantum Bet: Infrastructure Picks & Shovels

Bluefors (dilution fridges — not listed, watch for IPO), **Keysight** (control instruments), **Oxford Instruments** (cryogenics), **Linde** (helium supply), **FormFactor** (cryo probing). These win regardless of which qubit modality prevails. Every quantum computer needs cooling, control signals, testing, and helium. This is the "ASML strategy" applied to quantum — bet on the enablers, not the chip designers.

Quantum Optionality on Profitable Businesses

Alphabet (Willow chip breakthrough), **Microsoft** (Azure Quantum + topological), **IBM** (Qiskit + 1000+ qubits), **Honeywell** (54% Quantinuum owner), **Amazon** (Braket platform). These pass the 5-year test (\$63): you're buying profitable businesses where quantum is free optionality. If quantum works, massive upside. If it takes 20 more years, you still own Google/Microsoft/Amazon.

Post-Quantum Security: The Near-Term Revenue Play

"Harvest now, decrypt later" attacks make post-quantum cryptography (PQC) an **immediate** spending category — not a future one. NIST PQC standards are finalized. Every TLS certificate, VPN, HSM, and firewall needs migration. **Thales** (HSMs), **Palo Alto** (firewalls), **SandboxAQ** (crypto audit — private). PQC migration is a multi-year, multi-billion revenue wave that starts now, not when quantum computers break RSA.

Warning: Pure-Play Quantum Cash Burn

IonQ (~3.5yr runway), **Rigetti** (~2.5yr), **D-Wave** (~1.2yr), **QUBT** (minimal). Most pure-play quantum companies are burning \$80-120M/year with \$10-45M revenue. Dilution is the primary risk — share count has doubled or tripled for most since IPO. Only size positions you can afford to lose completely. D-Wave's runway is critical — may need capital raise imminently.

Hidden Champions Worth Tracking

Lasertec-equivalents of quantum: Disco/VAT/SCREEN have no quantum parallel yet because the industry is too small. But watch: **Classiq** (quantum EDA — if quantum scales, circuit design tools become essential), **ID Quantique** (QRNG already in Samsung phones — invest via SK Telecom), **Multiverse Computing** (Spanish quantum software for finance — relevant for local market).

Spanish Tax & Access Considerations (\$33–\$34)

Most pure-play quantum stocks are US-listed SPACs — no UCITS equivalent exists for quantum-specific ETFs. **Defiance Quantum ETF (QTUM)** and **Global X Quantum ETF** are US-listed (no traspaso). For tax-efficient access: use broad tech UCITS ETFs that include GOOGL/MSFT/IBM, and add individual quantum stocks as satellite positions through a broker with Spanish fiscal reporting. Telefónica (TEF)

	is the only Spanish-listed company with meaningful quantum exposure.
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Quantum Computing ETFs & Investment Vehicles

QTUM — Defiance Quantum ETF

Broadest quantum/ML ETF. Heavy in classical AI companies with quantum exposure.

AUM

~\$800M

Expense Ratio

0.40%

Holdings

~70

Top Holdings

NVDA, MSFT, IBM, IONQ

IONQ / RGTI / QBTS / D-Wave

Direct pure-play exposure. High volatility, high risk. Position sizing is everything.

Type

Individual stocks

Risk

Very high (pre-revenue)

Sizing

Max 1-2% of portfolio each

Horizon

10+ years minimum

Picks & Shovels Portfolio

KEYS + OXIG.L + LIN + FORM + HON — profitable companies with quantum tailwinds.

Type

Custom basket

Risk

Moderate (profitable cos.)

Quantum Purity

Low (5-15% of rev)

5-Year Test

Passes (\$63)

Big Tech Quantum Optionality

GOOGL + MSFT + AMZN + IBM + HON — quantum is free upside on profitable cores.

Type

Custom basket

Risk

Low (mega-caps)

Quantum Purity

Very low (<1% rev)

40-Year Thesis

Strongest approach

Suggested Quantum Allocation Architecture

Core (60% of quantum allocation) Big Tech optionality: Alphabet, Microsoft, Amazon — quantum programs backed by \$100B+ cash reserves. Honeywell for Quantinuum exposure. IBM if conviction in Qiskit ecosystem. These are core positions (\$16) — 3x evidence to rotate out. You're not betting on quantum working; you're betting on tech companies that happen to also be quantum leaders.	Satellite (40% of quantum allocation) Picks & shovels (20%): Keysight, Oxford Instruments, Linde, FormFactor. PQC security (10%): Thales, Palo Alto Networks. Pure-play (10%): IonQ (best funded), PsiQuantum (if IPO'd), D-Wave (smallest position — survival risk). Maximum 1-2% of total portfolio per pure-play. Satellite allows faster rotation (\$16). Rebalance as cash runway data changes quarterly.
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